PILOT

Guiding Long-Horizon World-Model Planning with Visual Subgoals and Policy Priors

68.8%

average real-world task success

4 tasks, 20 trials each

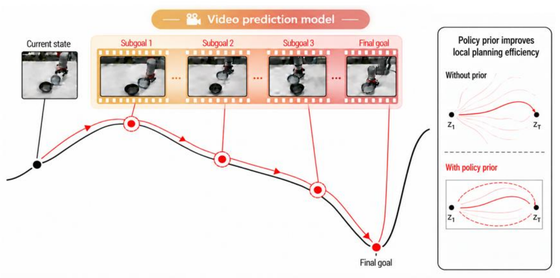
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* Equal contribution

MOTIVATION

Why PILOT works



Visual subgoals expose intermediate progress. The policy prior focuses local action search.

CORE IDEA

Two sources of structure

1 Visual temporal abstraction

Wan2.1 predicts the whole visual trajectory. Sampled keyframes become interpretable visual subgoals for local planning.

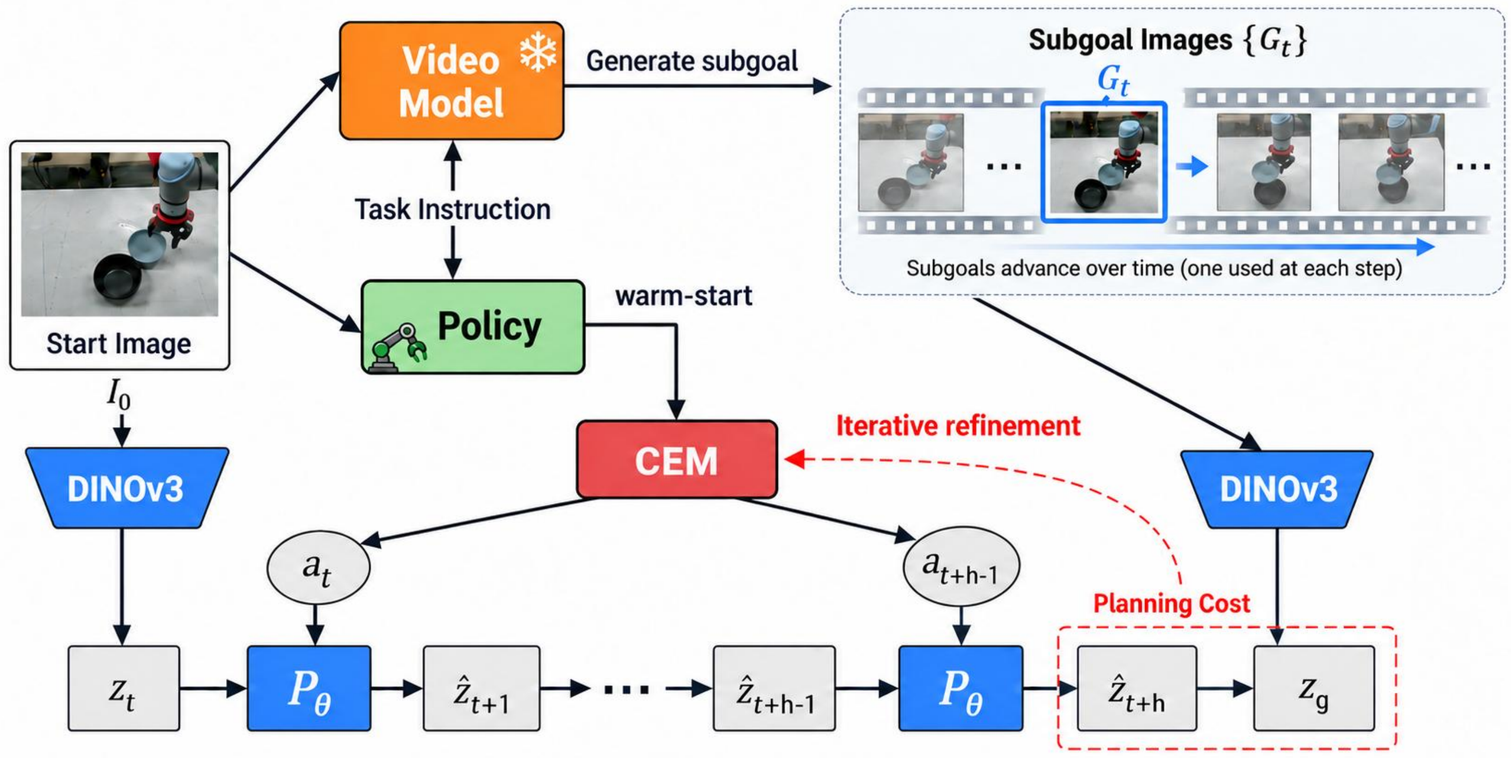
2 Behavior-aware local search

A policy centers CEM on plausible action chunks. JEPA-WM evaluates and refines them online.

Subgoals specify where to go. The policy prior specifies where to search.

METHOD

PILOT at a glance



The video model generates subgoals, the policy warm-starts CEM, and the action-conditioned world model evaluates and refines candidate actions online.

Receding-horizon planning loop

1 Generate
Predict visual subgoals

2 Initialize
Warm start CEM by policy

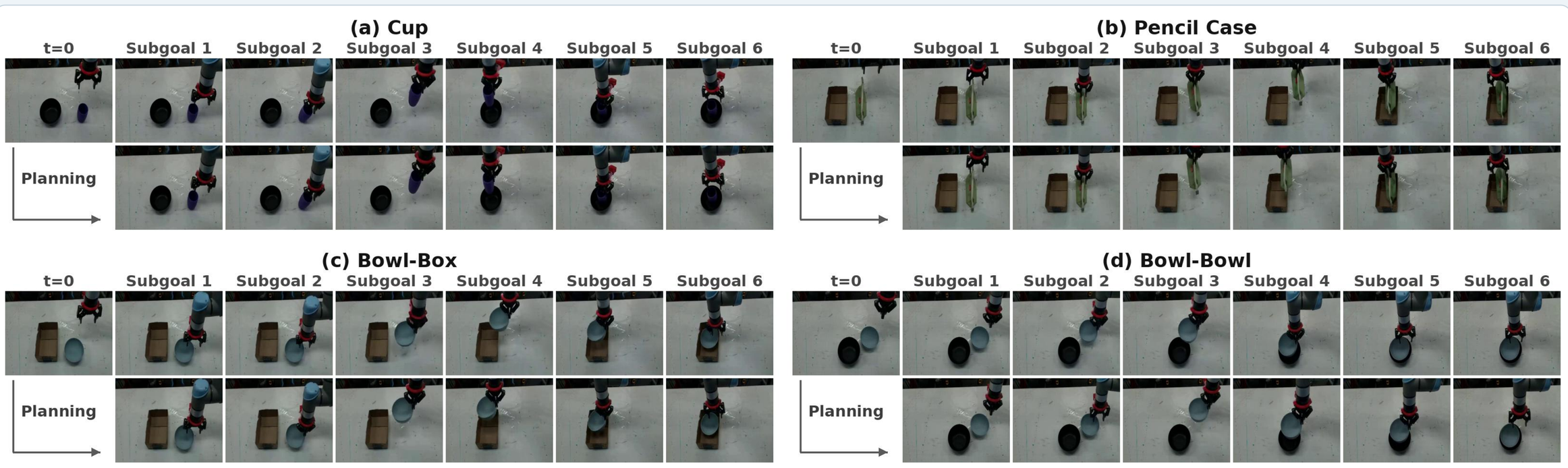
3 Optimize
CEM minimizes latent goal cost

4 Execute
Apply action, observe and replan

EXPERIMENTS

Real-world manipulation results

predicted subgoals above, executed robot states below



68.8%
PILOT
Cup 11/20 Pencil Case 13/20
Bowl-Box 13/20 Bowl-Bowl 18/20

55.0%
Without policy prior
Generic CEM initialization
13.8 pp below full PILOT

13.8%
Policy only
Direct execution of the learned policy

0.0%
Flat world-model planning
Direct planning toward the final goal image

TAKEAWAY

Temporal subgoals make the long task tractable. Policy-guided CEM makes each local search faster and more reliable.

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